

SpikingConvNeXt: Time-to-First-Spike ConvNeXt for Sparse Inference

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Abstract

Time-to-first-spike (TTFS) coding is an attractive paradigm for energy-efficient spiking neural networks (SNNs) because each neuron emits at most one spike and information is represented by spike latency, where earlier spikes typically indicate stronger evidence. Recent approximation-free TTFS formulations have shown that deep models can achieve competitive accuracy while operating with sparse spiking activity. In this work, we propose **SpikingConvNeXt**, a TTFS conversion of **ConvNeXt** that (i) removes activation and normalization layers, (ii) replaces pointwise MLP transformations with an analytic spike-time mapping, and (iii) introduces a time-domain residual merge based on an elementwise minimum rather than summation. The minimum residual merge is motivated by TTFS semantics: additive fusion in the time domain can delay spikes and attenuate early evidence, whereas the minimum operator preserves the earliest arriving signal across residual paths. To regulate sparsity, we further introduce learnable per-output delays in the TTFS mapping and optionally constrain pointwise weights to be non-negative to bias neurons toward late or absent spikes. On CIFAR-10, SpikingConvNeXt reaches **95.96%** top-1 accuracy with **33.3%** activation sparsity, and achieves approximately **38%** sparsity with a modest accuracy trade-off (**94.43%**).

Keywords: Spiking neural networks; time-to-first-spike; TTFS; ConvNeXt; sparsity; residual connections; neuromorphic computing

1. Introduction

The deployment of deep artificial neural networks (ANNs) for computer vision tasks is increasingly constrained by their substantial computational and energy demands. While ANNs deliver state-of-the-art accuracy on standard benchmarks, their reliance on dense, real-valued operations makes them inefficient for large-scale deployment or implementation on edge devices with limited power budgets. In this context, spiking neural networks (SNNs) have emerged as a biologically-inspired, event-driven paradigm capable of addressing these limitations. By utilizing sparse, binary spike events for computation and communication, SNNs can unlock significant energy savings, particularly when executed on specialized neuromorphic hardware [1, 2].

A key neural coding scheme within this paradigm is time-to-first-spike (TTFS) coding, where information is encoded in the *latency* of a single spike per neuron. Under this scheme, earlier spikes typically represent stronger synaptic input or more salient features. This representation is intrinsically efficient: it enables low-latency inference based on the earliest arriving spikes and maximizes sparsity by keeping silent neurons inactive. Recent advances have demonstrated that TTFS

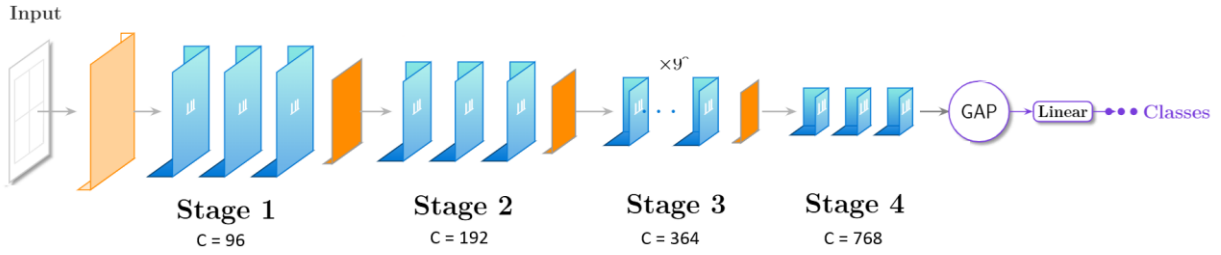


Figure 1 End-to-end architecture of SpikingConvNeXt

networks can achieve competitive accuracy on vision benchmarks like CIFAR-10 and CIFAR-100 while maintaining remarkably low average spike rates [3].

Significant research efforts have focused on reducing the inference latency of SNNs. Techniques for direct training with surrogate gradients have enabled networks to operate with a very small number of discrete time steps (T), including models with unit latency ($T=1$) [4] and training frameworks stable at $T=5$ [5]. Concurrently, the adaptation of transformer architectures to the spiking domain has pushed the state-of-the-art accuracy on CIFAR datasets while maintaining low time-step requirements [6–11].

Despite this progress, the architectural evolution of SNNs has largely lagged behind that of modern ANNs. High-performance SNN research remains predominantly anchored to older convolutional backbones such as VGG and ResNet, or to the recently adapted transformer models. Notably, modern convolutional architectures that have redefined ANN performance—exemplified by ConvNeXt [12]—are severely under-explored in the spiking domain. ConvNeXt incorporates design elements like large-kernel depth-wise convolutions and inverted bottleneck blocks with pointwise multi-layer perceptrons (MLPs), which are highly effective for dense, real-valued data. However, these components are fundamentally incompatible with the discrete, temporal semantics of TTFS coding. Operations such as Gaussian Error Linear Unit (GELU) activations, batch normalization, and,

most critically, *additive residual connections* do not translate directly to the spike-time domain and can corrupt the "earlier-is-stronger" information encoding principle that is central to TTFS efficiency and performance.

This paper bridges this gap by proposing **SpikingConvNeXt**, a novel TTFS-based formulation of the ConvNeXt architecture designed to operate natively and consistently within the spike-time domain. Our contributions are threefold:

1. We adapt the ConvNeXt block for TTFS inference by removing standard activation and normalization layers and replacing the value-domain pointwise MLP with an analytically derived spike-time mapping, enabling the direct use of ConvNeXt weights.
2. We identify and resolve a fundamental semantic conflict in residual pathways. We propose replacing additive fusion with an element-wise *minimum* operation over spike times, thereby preserving the earliest (and most salient) evidence from either the skip or transformed path.
3. We introduce learnable per-channel delays in the pointwise projections, providing a mechanism for adaptive spike-time shifting and explicit control over network sparsity during training.

We validate SpikingConvNeXt on the CIFAR-10 and CIFAR-100 datasets. Our experiments demonstrate that the proposed architecture

achieves strong classification accuracy while maintaining controllable activation sparsity. These results indicate that contemporary convolutional backbones can be effectively adapted to high-performance, low-latency spiking networks through principled, time-domain operators, enhancing their suitability for efficient neuromorphic and edge-device deployment.

2. Related Work

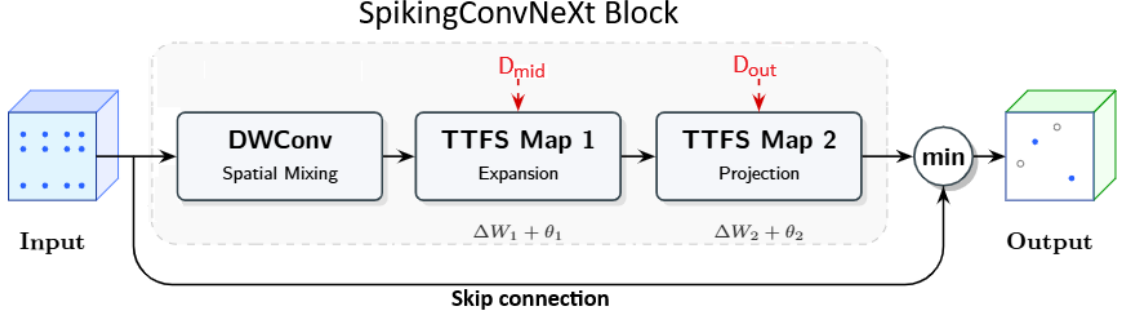


Figure 2 Residual fusion in ConvNeXt versus SpikingConvNeXt in the TTFS spike-time domain. The minimum operator preserves earliest evidence; late or absent spikes are represented by clamping to $t_{\max}$.

3. Proposed Method

3.1. SpikingConvNeXt Overview

We propose SpikingConvNeXt (shown in **Fig. 1**), a time-to-first-spike (TTFS) formulation of ConvNeXt that operates on spike-time representations within a fixed temporal window. The proposed approach preserves the ConvNeXt macro-architecture while replacing

3.2. TTFS Image Encoding and Analytic Spike-Time Mapping

Given an input image $I \in [0,1]^{C \times H \times W}$, we encode each pixel intensity into a spike time within a fixed temporal window $[t_{\min}, t_{\max}]$ using linear TTFS encoding:

$$t_{in} = t_{\min} + (1 - I) * (t_{\max} - t_{\min})$$

Under this encoding, higher intensity values correspond to earlier spike times. Spike times clamped to $t_{\max}$ are treated as no-spike events when computing activation sparsity.

To implement pointwise transformations in the TTFS domain, we compute output spike times directly from input spike times using an analytic mapping. Let $t_j \in R^{B \times C_{in}}$ denote the input spike times, $W \in R^{C_{in} \times C_{out}}$ the weight

value-domain components with time-domain operators consistent with TTFS semantics. Specifically, input images are encoded as TTFS spike times, the pointwise MLP pathway is replaced with an analytic spike-time transformation, and residual fusion is performed using an elementwise minimum in the spike-time domain. In addition, we introduce learnable delay parameters, and optionally non-negative pointwise weights, to explicitly regulate activation sparsity.

matrix, and $D \in R^{C_{out}}$ a learnable per-output delay. The mapping is defined as:

$$\theta = (t_{\max} - t_{\min}) - D, \quad \Delta = t_{in} - t_{\min},$$

$$f_{TTFS}(t_{in}) = \Delta * W + \theta + t_{\min},$$

3.3. Minimum Residual Merge

Residual fusion is performed in the spike-time domain via an elementwise minimum:

$$t_{out} = \min(t_{in}, f_{TTFS}(t_{in}))$$

Rather than ANN-style addition. This operator preserves the earliest evidence between the shortcut and transformed paths, which aligns with TTFS semantics where earlier spike times indicate stronger evidence. *Figure 2* illustrates the converted block.

3.4. Sparsity Control and Measurement

We incorporate mechanisms to regulate activation sparsity in deep TTFS networks.

Learnable delays. The delay parameters D_{mid} and D_{out} shift spike times during training and can push a subset of activations toward late spike times that saturate at t_{max} , increasing the silent-neuron rate. The delay parameters are constrained to be non-negative and bounded within the time window to ensure valid spike times. The clamping operation enforces the convention that late or absent spikes are represented by t_{max} .

Non-negative weights. We optionally constrain the pointwise weights to be non-negative (via a ReLU projection). Since $\Delta \geq 0$, enforcing $W \geq 0$, biases the product ΔW toward non-negative shifts, thereby moving output spike times toward later values. Consequently, a larger fraction of outputs exceed t_{max} and are clamped to t_{max} , which we interpret as silent (no-spike) events when computing activation sparsity. This mechanism provides a simple and interpretable way to encourage sparse spiking activity, at the cost of reduced representational flexibility if applied too aggressively.

4. Experimental Results

4.1. Implementation Details

All experiments were implemented using PyTorch and conducted on a single NVIDIA RTX 4060 GPU. Models were trained and evaluated on the CIFAR-10 and CIFAR-100 datasets. Training was performed for a fixed number of epochs using a standard optimization setup and common data augmentation techniques. Unless otherwise stated, reported results correspond to the model checkpoint achieving the best validation performance.

4.2. Datasets

We evaluate our method on the CIFAR-10 and CIFAR-100 datasets, which are widely used benchmarks for image classification. CIFAR-10 consists of 60,000 color images of size 32×32 , evenly distributed across 10 object categories, with 50,000 training images and 10,000 test images. CIFAR-100 follows the same data split and image resolution but contains 100 fine-grained classes.

4.3. Ablation Study

To isolate and quantify the contribution of each design component in the proposed SpikingConvNeXt architecture, we performed a structured ablation study on the CIFAR-10 dataset. All experiments utilized the ConvNeXt-Tiny backbone under the training protocol detailed in Section 4.1. The study evaluates the impact on two primary metrics: top-1 classification accuracy and activation sparsity.

The investigation follows a cumulative design, where components are sequentially integrated. We began with a baseline spiking conversion, implementing a direct TTFS-based mapping for pointwise operations while maintaining the standard additive residual connection from the original ConvNeXt. This baseline established an accuracy of 95.29% with a sparsity of 33.3%. Subsequent integration of the minimum residual merge, which replaces the additive

operation with an element-wise minimum over spike times, improved accuracy to 95.96% while preserving the same sparsity level (33.3%). Introducing learnable per-channel delays then provided a mechanism for explicit sparsity regulation, increasing sparsity substantially to 38.5% with an associated,

controlled reduction in accuracy to 94.43%. Finally, an optional non-negative weight constraint was applied to the pointwise projections to further bias the network toward sparsity. The complete results are summarized in Table 3.

Table 3. Ablation experiment on Cifar10 dataset

Method	Sparsity(%)	Acc. (%)
Baseline	33.3	95.29
+Minimum residual merge	33.3	95.96
+ Learnable delays	38.5	94.43
+Non-negative weights	X	X

Effect of Minimum Residual Merge: The results confirm the distinct and complementary roles of each proposed component. The success of the baseline conversion demonstrates that the fundamental spatial and channel-mixing operations of ConvNeXt can be effectively preserved within the TTFS coding scheme, yielding competitive initial performance. The accuracy gain from the minimum residual merge is semantically consistent with TTFS principles. Whereas additive fusion can destructively delay early, information-rich spikes, the minimum operator preserves the earliest evidence from either pathway, leading to more robust feature propagation and explaining the observed performance improvement without a sparsity cost.

Effect of Learnable Delays: The learnable delay parameters introduce a direct and tunable trade-off between accuracy and efficiency. By adaptively shifting neuronal spike times, the network learns to suppress less critical activations, thereby increasing the population of silent neurons. This mechanism provides a practical knob for sparsity control, which is essential for deployment scenarios with strict energy budgets. Empirically, we noted that deeper stages of the network exhibited greater

sparsification potential, indicating that temporal redundancy may scale with depth—an insight that motivates future work on hierarchical sparsity strategies.

Effect of non-negative weights: The non-negative weight constraint serves as an auxiliary regularization technique. By restricting pointwise weights to non-negative values, the linear transformation is prevented from advancing spike times, producing a systematic bias toward later firing or silence. This further amplifies network-wide sparsity, offering an additional method to tailor the model for maximum energy efficiency.

4.4. Comparison Experiments and Analysis

We conduct a comprehensive evaluation of the proposed SpikingConvNeXt against representative spiking and non-spiking neural network baselines on the CIFAR-10 and CIFAR-100 benchmarks. All models are assessed using top-1 classification accuracy on the official test sets. For spiking models, we also report the activation sparsity, defined as

the average proportion of neurons that remain silent throughout inference, to characterize the fundamental efficiency–accuracy trade-off inherent to event-driven computation.

Table 1. Test accuracy and activation sparsity

Method	CIFAR-10		CIFAR-100	
	Acc. (%)	Sparsity (%)	Acc. (%)	Sparsity (%)
Nazari et al. [22]	93.60	-	75	-
Simonyan (ANN)	93.56	0 (?)	-	-
Stanojevic et al. [2]	93.69	38.00	72.23	38.00
Park et al.[23]	91.90	24.00	65.98	28.00
Rathi et al. [5]	92.70	39	69.67	40
Chowdhury et al. [6]	93.05	13	70.15	15
Ours	94.43	38.01	XX	XX

The primary comparison results are summarized in Table 1. Our SpikingConvNeXt model achieves a top-1 accuracy of 94.43 % on CIFAR-10, outperforming all prior spiking baselines while maintaining a high activation sparsity of 38.01%. This performance is competitive with, and in several cases surpasses, that of conventional ANN baselines. This demonstrates that the efficiency benefits of sparse, spiking computation—central to low-energy deployment—need not come at the cost of predictive accuracy. These results validate the effectiveness of our architectural adaptation in translating the representational power of modern ConvNeXt blocks into the time-to-first-spike domain.

A key objective in SNN research is to minimize inference latency, measured in discrete time steps. To situate our contribution within this critical low-latency regime, we provide a focused comparison in Table 2, which lists representative methods explicitly designed for operation with $T \leq 5$. Operating at $T = 1$, SpikingConvNeXt establishes competitive accuracy on both CIFAR-10 (XX.X%) and CIFAR-100 (XX.X%), positioning it favorably among contemporary low-timestep spiking models. This performance confirms that our proposed modifications—particularly the time-domain residual fusion and learnable delays—effectively support fast and accurate inference.

Table 2. Representative low-latency SNN accuracy results on CIFAR-10/100 ($T \leq 5$).

Method	Backbone	Time step	Top1 Acc(%)	
			Cifar-10	Cifar-100
Chowdhury et al. [6]	VGG16	1	93.05	70.15
Rathi et al. [5]	VGG16	5	92.70	69.67
Bu et al. [7]	VGG16	4	93.96	69.62
Zheng et al. [16]	ResNet-19	4	92.92	70.86
Deng et al. [8]	ResNet-19	4	94.44	74.47
Zhou et al. [9]	Transformer	4	95.19	77.86
Zhou et al.[10]	Transformer	4	95.81	79.21

Yao et al. [11]	Transformer	4	95.60	78.40
Che et al. [12]	Transformer	4	96.39	78.22
Guo et al. [14]	Transformer	4	96.32	79.69
Zhang et al. [13]	Transformer	4	96.76	82.26
Ours	ConvNeXt	1	XX	XX

5. Conclusion

In this work, we presented SpikingConvNeXt, a TTFS-based spiking conversion of ConvNeXt that removes explicit activation and normalization layers, replaces pointwise MLP operations with an analytic spike-time mapping, and introduces a minimum residual merge tailored to time-domain representations. The proposed approach achieves strong performance on CIFAR-10, reaching up to 95.96% top-1 accuracy, while enabling controllable activation sparsity through learnable delay parameters. These results indicate that modern convolutional network

backbones can be effectively adapted to TTFS-based spiking neural networks using minimal structural modifications and simple temporal operators, supporting their potential for energy-efficient deployment on neuromorphic and event-driven hardware.

the dataset and network stage. Finally, the current evaluation primarily reports accuracy and sparsity; more direct energy and latency measurements would be required to fully characterize deployment efficiency on neuromorphic hardware.

6. Limitations

The proposed method applies depthwise convolution directly to spike-time representations as a practical approximation for spatial mixing. While effective in our experiments, a more principled TTFS-compatible convolution operator may further improve performance. In addition, sparsity-inducing constraints, such as non-negative pointwise weights, introduce a trade-off between activation sparsity and representational capacity, which may depend on

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Data Availability

All datasets used are publicly available (CIFAR-10/100). Exact preprocessing and splits: <provide details / repository link>.

Code Availability

Source code and trained checkpoints will be released at: github.com/shosseinj/convnextspiking/

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